What is Artificial Intelligence?

Artificial Intelligence (AI) is the **broader field** that aims to make machines act **smart**—like humans.

Something that can **think, learn, move, talk, and/or make decisions** is called **an intelligent system**.

For example:

- **A human** is an intelligent system—we think, learn from experience, and make choices every day.
- **An AI chatbot** can also be an intelligent system—it learns from data, understands your questions, and gives helpful answers.
- **Self-driving cars** are intelligent systems too—they see what's around them, learn from the environment, and decide when to stop, turn, or accelerate.

So, anything—whether it's a person, computer program, or robot—that can **process information, learn from it, and take smart actions** is showing intelligence.

In short:

> **Artificial = man-made**
> **Intelligence = the ability to think, learn, and decide/make decisions**
> **Artificial Intelligence = Man-made system that can think, learn, and make decisions**

Example Video on Robotics

https://www.youtube.com/watch?v=FVJy4B74ucs

Areas of Artificial Intelligence

1. **Machine Learning (ML)**
   Teaching computers to **learn from data** and make predictions or decisions.

   Example: Recommending movies on Netflix.

---

2. **Natural Language Processing (NLP)**
   Helping machines **understand human language** (text or speech).

   Example: Chatbots, Google Translate, Siri.

3. **Computer Vision**
   Teaching computers to **see and understand images or videos**. Use **Deep Learning Techniques**

   Example: Face recognition, self-driving cars, medical image analysis.

---

4. **Robotics**
   Making smart robots that can **move, sense, and interact** with their environment.

   Example: Delivery robots, robotic arms in factories.

---

5. **Expert Systems**
   Rule-based systems that use expert knowledge to **solve specific problems**.

   Example: Virtual doctor, Medical diagnosis tools, legal decision assistants.

---

6. **Speech Recognition**
   Turning **spoken words into text** so computers can understand voice commands.

   Example: “Hey Google” or voice typing.

In Simple Terms:

AI is about making machines smart — so they can think, learn, see, talk, move, and make decisions.

In the Next Lessons, we look at Some of these AI Areas.

SpeechToText

Check https://github.com/modcomlearning/SokoGardenApp Step 6

Natural Language Processing

Robotics

Natural Language Processing (NLP)

What is NLP?

Natural Language Processing (NLP) is a field of Artificial Intelligence (AI) that focuses on enabling computers to **understand, interpret, and respond** to **human language** in a way that is both meaningful and useful.

Applications of NLP:

- **Chatbots** and virtual assistants (e.g., Siri, Alexa)
- **Language translation** (e.g., Google Translate)
- **Speech recognition** (e.g., voice typing)

Example of a Chatbot

A chatbot is an AI program that simulates human conversation.
Example:

> User: *"What courses do you offer?"*
> Chatbot: *"We offer programs in Android development and Full Stack Development."*

Practical

In this practical, we create a Chatbot using NLTK Python Library .

Install NLTK.

pip install nltk

Introduction of Regex – Regular Expressions.

Regular expressions (regex) are powerful tools used to search, match, and manipulate text based on specific patterns. Example check if a character exist in a sentence. Or @ sign exist in an email.

Check Regex PDF.